



PROJECT-BASED LEARNING

PHASE-I EVALUATION

Detective Mystery: The Hidden Truth

Team ID: DSCPP-III-2026-T152

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026-27



Team & Mentor Details

Team Information

Team ID: T152

Semester: 3rd

Section: DS2,F,J,K

Domain: *Game Development*

Team Members

1. *Karan Badhani : 2510360037*
2. *Khushi Singal : 2510360039*
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Mentor

Prof. Dr. Jyoti Agarwal

Interactions completed: 3



Problem Statement & Motivation

Problem Statement

Traditional mystery-solving methods lack an interactive way to investigate clues, connect evidence and identify the culprit. This project develops a C++ & DSA-based detective mystery game for an engaging investigation experience.

Motivation

- ▶ *Make mystery-solving interactive and engaging*
- ▶ *Improve logical thinking & problem-solving*
- ▶ *Make learning fun and interesting*

Target Users / Stakeholders

- ▶ *Primary users: Students & game players*
- ▶ *Secondary stakeholders: Teachers / Faculty*
- ▶ *Expected benefit: Interactive learning & practical DSA experience*



Objectives

Project Objectives

1. *Develop an interactive 2D detective mystery game for practical investigation.*
2. *Implement core game features: exploration, clues, suspects, evidence analysis, and final verdict.*
3. *Integrate OOP and DSA concepts using C++ in the game.*
4. *Evaluate game functionality through test cases for navigation, clues, investigation, and final deduction.*
5. *Demonstrate practical use of OOP and DSA in an interactive game system.*

Key Objective: *Create an interactive detective game where players solve mysteries by finding and connecting clues, while demonstrating OOP and DSA concepts.*



Proposed Solution

Proposed Approach

1. *Player Input and Case Setup*
2. *Explore Locations and Collect Clues*
3. *Analyze Evidence and Interrogate suspects*
4. *Apply DSA for Investigation*
5. *Final Verdict and Scoring*

Key Features

- ▶ *Interactive Mystery Investigation*
- ▶ *Graph-Based Location Exploration*
- ▶ *Clue and Evidence Management*
- ▶ *Suspect Interrogation System*
- ▶ *Scoring and Multiple Endings*



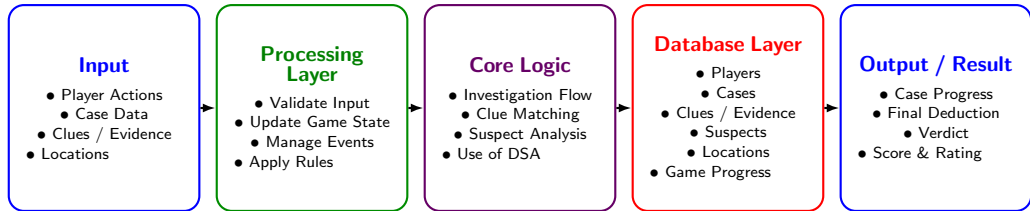
Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	<i>Arrays, Linked Lists, Trees, Graphs, Hashing</i>	<i>Investigation & Clue Management</i>
OOPs with C++	<i>Classes, Inheritance, Polymorphism, Encapsulation</i>	<i>Investigation engine & Game</i>
Operating Systems	—	—
DBMS	—	—

Requirement: OOP concepts create and manage game entities such as suspects, clues, evidence and locations while DSA organizes and processes them for investigation, navigation and clue analysis.



System Architecture / Workflow



Major Components

- ▶ *Player and Game Management*
- ▶ *Investigation and Location Management*
- ▶ *Clue, Evidence and Suspect Management*

Data / Control Flow

- ▶ *Player input and case data*
- ▶ *Clues and evidence processed through game logic*
- ▶ *Game result generated*
- ▶ *Expected output: Final Verdict & Score*



Technology Stack & Methodology

Technology Stack

- ▶ Programming: *C++*
- ▶ Database: *file handling*
- ▶ Tools / Framework: *SFML (Simple and Fast Multimedia Library)*
- ▶ IDE: *Visual Studio Code*
- ▶ Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

Justify the selection of technologies and methodology.



Initial Progress & Team Contribution

Progress Achieved

- ▶ *Detective mystery game storyline and case structure finalized.*
- ▶ *OOP classes such as Player, Suspect, Clue, Evidence and Location identified.*
- ▶ *Mapped Graph, Stack, Queue, Hashing, Tree, Searching & Sorting to gameplay.*
- ▶ *System Architecture ,Workflow prepared,C++ + SFML technology stack finalized*

Individual Contribution

Member	Role	Contribution
M1	Lead	Storyline, planning & system design
M2	Developer	Player movement & game mechanics
M3	DSA developer	DSA integration & investigation logic
M4	Documentation/UI	Documentation, UI & testing



Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- ▶ *Functional 2D detective game prototype*
- ▶ *Practical implementation of OOP DSA*
- ▶ *Expandable for future cases and levels*

References

1. *SFML Official Documentation*
2. *C++ Reference (cppreference)*
3. *GeeksforGeeks – Data Structures*
4. *Microsoft Learn – C++ Documentation*

Thank You
Questions & Suggestions